SimWorld Research White Paper

A reproducible reduced model and a research program for emergence across physical and social scales

29 October 2026 | Version 1.1 | Physics build 0.1 | All 17 detailed phase plans

Abstract

SimWorld proposes a continuous simulated history in which higher-level structures arise from local interactions, extending from a hot physical seed through chemistry, planetary environments, life, technology and institutions. This paper formalizes that ambition as a multiscale research program and assesses the executable baseline against its design documents. The present implementation is a small stochastic model of thermal populations, effective nucleon formation, light-nucleus reactions and electron capture. It includes event logging, deterministic re-execution, canonical state hashes and restricted skipping of quiet cells. These are concrete foundations for reproducible simulation experiments.

The original 29 October 2026 audit found all 65 existing tests passing. Its eight-cell, 3,400-tick realization at seed 42 reproduced the archived summary and event log, and replay matched all 34 recorded hashes. Helium-4 mass fraction was 0.2276 and the reported ionized fraction 0.003457. Subsequent public CI passed the 65-test suite independently on Windows and Linux. These results establish the tested regression and replay properties; they do not establish precision cosmology or cross-platform bit identity.

The preserved diagnostics identify missing reverse-reaction reachability, an absorbing fully neutral atomic state, repeated random-stream prefixes, and a mismatch between prototype performance and production storage. Version 1.1 adds detailed profiles of every phase from 0 to 16, the current dependency ledger, and public repository and deployment evidence. Each profile states mechanisms, prerequisites, proposed validation and limits. The central question remains whether transparent effective models can preserve useful causal structure while supporting increasingly complex, unscripted behavior at affordable computational cost.

Keywords: multiscale simulation; emergence; reproducibility; stochastic kinetics; coarse-graining; artificial life; agent-based modeling.

Evidence date: 29 October 2026. Source baseline: commit 4134abd. This revision updates the research specification and project status; the original simulation measurements retain their original provenance. Editable paper sources and both current and historical source manifests accompany the publication.

Current project status

Version 1.1 records the public project at commit **4134abddc30940f455209c442afe68f442ebaa75** on 29 October 2026. It incorporates all seventeen detailed phase plans and the subsequent publication infrastructure. The executable physics baseline remains 0.1; the expanded plans do not represent implemented chemistry, planets, biology or society. [S8]

Area	Verified state at the source baseline
Implementation	19 Python modules; reduced stochastic micro-seed; no executable AST changes since the original audit after removing docstrings.
Planning	PHASE0 through PHASE16 available. Each is profiled in Appendix B; future mechanisms remain proposals.
Dependency ledger	25 entries, 22 open; S14, S17 and S21 record wording or interface-contract corrections.
Windows CI	65 passed in 136.61 seconds; Python 3.11.9, NumPy 2.4.6, pytest 9.1.1.
Ubuntu CI	65 passed in 112.61 seconds with the same pinned engine package versions.
Publication	Public repository, contribution/architecture guides, static GitHub Pages site, checksummed research downloads and automatic deployment.

Status table. Test durations are pytest output from the cited jobs, not controlled engine-throughput comparisons. Repository publication improves access and reproducibility; it does not validate unbuilt systems.

The independent platform checks expand the verification evidence beyond the original local run. They do not establish bit-identical state histories between operating systems, complete physical conservation, or correctness of the known reverse-reaction and stream-identity gaps. Those remain explicit research and implementation work.

The public records are the repository, successful CI run 40857373977, and Pages deployment 40857374197. The project site is enderpeer.github.io/simworld. Machine-readable publication_status.json preserves these identities and the plan fingerprints.

The revised research archive preserves the original audit evidence alongside the newer source snapshot. New profiles emphasize conserved ledgers, dependency contracts, approximation boundaries and falsifiable tests. The document version is 1.1; this is a publication revision, not a new physics build.

1 Research problem and scientific position

The project asks whether one causal history can connect microscopic interactions to macroscopic organization without scripting the order of discoveries or prescribing a final society. The roadmap emphasizes computed properties, regenerable caches, persistent history and observation-dependent allocation of detail. Its 17 phases supply a wide research scope, while BUILD0 specifies a deliberately narrow first executable slice. A scientifically useful account must preserve both the ambition and the limited domain of the evidence. [S1, S2]

The defensible organizing hypothesis is that higher-level models can retain selected consequences of lower-level dynamics through explicit interfaces. Such models necessarily choose state variables, effective rates, constitutive relations and learning rules. A property computed from a fitted formula is derived within that model, but it is not an independent first-principles prediction. Avoiding element names or runtime lookup tables does not remove calibration, structural priors or excluded mechanisms. SimWorld should disclose these choices as part of the model rather than treat them as violations of its purpose.

Relationship to prior research

Precision big-bang nucleosynthesis provides a benchmark for coupled weak and nuclear reaction calculations, including uncertainty treatment. The observational deuterium abundance provides an independent target with a scale far below the granularity of this baseline. Recombination research also demonstrates that expensive microscopic calculations can be compressed into effective rates while retaining a carefully defined physical domain. These precedents support derived caches and model reduction, provided their accuracy is measured. [1-3]

Heterogeneous multiscale methods couple coarse evolution to local microscopic calculations. They motivate the proposed hierarchy, but do not prove scale separation for the entire chain from particles to institutions. For stochastic reactions, Gillespie's algorithm supplies a reference sampler for a specified well-mixed kinetic model. For agents, the ODD protocol supplies a disciplined description of purpose, state, scheduling and submodels. Each addresses a distinct part of scientific reproducibility. [6-8]

Claims that the present evidence can support

The present contribution is an auditable reduced simulation and a structured research agenda. This paper makes no claim that SimWorld solves quantum field theory, predicts primordial abundances precisely, produces general chemistry, generates life, or explains human history. Those questions require additional models and experiments. The roadmap's unbuilt phases are evaluated as hypotheses about mechanisms and interfaces, not as achieved capabilities.

A meaningful emergence claim names the local rules, the measured higher-level pattern, the initialization and the interventions that distinguish competing explanations. Successful replay shows that an experiment can be repeated. Independent validation asks whether the experiment represents its intended physical or social system. A useful model can satisfy one criterion before it satisfies the other.

2 Evidence and audit methods

The evidence base consists of ROADMAP.md, all 17 detailed PHASE0.md through PHASE16.md plans, the BUILD0 contract, specification records, executable Python modules, tests, two prototype studies and archived runs. Version 1.1 also verifies repository publication and public CI at source commit 4134abd. Assertions were checked against executable paths rather than accepted from design prose alone. Literature selection is focused methodological context, not a systematic review. [S1-S8]

Evidence class	Material examined	Permitted interpretation
Design intent	ROADMAP and PHASE0 to PHASE16; BUILD0	Proposed scope, assumptions and acceptance criteria
Implemented mechanism	sim package and tests	Behavior supported by inspected code and executed checks
Archived measurement	M3 and M4 run records; spike reports	Historical observation with its own workload and environment
Verified status	Original diagnostics; local and public CI; repository and Pages records	Dated evidence with exact source and environment boundaries
External context	Primary research and methodological papers [1-10]	Benchmarks and methods; no endorsement of SimWorld

Table 1. Evidence categories prevent proposed mechanisms and prototype measurements from being reported as capabilities of the main engine.

The original local audit used Python 3.11.9, NumPy 2.4.6 and pytest 9.1.1: all 65 tests passed in 123.64 seconds. A later local publication check passed in 121.26 seconds. At source commit 4134abd, public CI passed the same 65 tests on Windows in 136.61 seconds and Ubuntu in 112.61 seconds, using those pinned engine versions. These pytest durations are observations of their environments, not controlled performance comparisons. [S5, S7, S8]

Verification compared complete summary objects, saved hash sequences and SHA-256 digests of decompressed event logs. Comparing decompressed logs avoids treating gzip container metadata as simulated history. The original audit run and replay took approximately 5.55 seconds each while the test suite was also active. These are single observations under concurrent load, not controlled performance benchmarks or estimates of future scalability.

Three additional diagnostic cases tested mechanisms outside the existing acceptance suite: dissociation from a product-only nuclear population, evolution from a fully neutral atomic population, and repeated calls to the same random-stream coordinates with different payloads. The exact diagnostic program and output accompany the paper. These checks expose limitations even though the current tests pass. They do not replace a comprehensive statistical or physical validation campaign. [S7]

Since the first audit, six modules received explanatory comment and docstring corrections. Independent comparison of all 19 modules found no executable abstract-syntax-tree changes after docstrings were excluded. This revision changes the paper and publication materials, not physics_build 0.1. Source hashes and exact commit identities distinguish the revised documentation from the historical audit snapshot. [S8]

3 The implemented reduced physical model

State and expansion

The executable world contains cell temperatures, quant records, integer nuclide populations, captured-electron counts and summary statistics. QuantStore uses Python dictionaries of per-quant lists and per-cell species pools, with NumPy count arrays. It does not implement the fully array-based structure-of-arrays layout described in the frozen D1 decision. Nuclide and atomic states are aggregate counts. The current cell system has a shared analytic temperature and does not evolve spatial fluid transport, gravity or a three-dimensional universe. [S3, S4]

$$t_n = 10^{-5}\,(1.01)^n \text{ s}, \quad a(t) = (t / t_0)^{1/2}, \quad T(t) = 100 / a(t) \text{ MeV}. \quad (1)$$

Equation 1 prescribes radiation-like cooling throughout the run. It is a background schedule rather than a Friedmann evolution solved from simulated energy density. The five effective field entries carry specified masses, charges, spins and coupling labels. Thermal target counts are rounded functions proportional to T cubed with a piecewise massive-particle suppression factor, scaled by the calibration volume. Thus the existence and properties of these field entries are inputs. [S4: constants, fields, lattice and condense]

Effective nuclei and reaction discovery

Weak flavor conversion supplies a frozen neutron probability. Below the confinement threshold, available quanta are consumed in triples, and neutron/proton labels are sampled from that probability. The consumed triple need not carry the sampled label's exact flavor content; the project records this as sea completion. Nuclear binding then uses a fitted liquid-drop expression, with nucleons assigned zero binding:

$$B(Z,A) = a_v A - a_s A^{2/3} - a_c Z(Z-1)/A^{1/3} - a_a(A-2Z)^2/A + a_p \delta(Z,A) A^{-3/4}. \quad (2)$$

The pairing factor is zero for odd A and has opposite signs for even-A systems with even or odd Z. Five fitted coefficients are published in constants.py. The network considers a bounded set with Z at most 4 and A at most 12. A candidate sum must have positive computed reaction energy, but additional explicit rules reject mass-5 nuclei and products with neutron-proton imbalance greater than one. These structural restrictions are model assumptions; the resulting stability pattern is not wholly emergent from the binding formula. [S4: nucbind, bind and network]

Forward reactions use population products, a fitted rate scale and charged-particle suppression. Rates are converted to capped per-tick probabilities, and the fixed update order limits which events can occur in each tick. This is not Gillespie sampling, and its rates cannot be interpreted as continuous-time physical rates without a consistent time mapping and convergence study. The reverse-channel implementation has additional limitations discussed in Section 6. [8, S4]

4 Atoms observation and reproducible execution

Electron capture

The atomic module computes the hydrogenic scale $I = \alpha_{em}^2 m_e/2$, approximately 13.61 eV for the configured constants. It stores one captured-electron count per cell and nuclide, using a spread-filling rule to reconstruct capture levels. Free electrons are the difference between total nuclear charge and captured electrons. This enforces a charge-accounting identity; it does not simulate electrons as orbital wavefunctions or a dynamical photon bath. Capture is disabled above $T_{gate} = 3I$. [S4: atoms]

The comparison function is a model-specific Saha-like solution. Writing $L = \exp(I/T)/\lambda^3$ and $K = NL$, it returns the positive solution to $Kx^2 + x - 1 = 0$:

$$x_S = [\mathrm{sqrt}(1 + 4K) - 1] / (2K). \quad (3)$$

Here N is the hydrogen count per cell, $\lambda^3 = (\mu T\ \mathrm{SAHA_PHI_E})^{3/2}$, and $\mu = m_e A/(A+1)$, with A = 1 for hydrogen. The phase-space factor is calibrated in engine units. Equation 3 is an internal reference, not independent cosmological data. Fully captured species are excluded from updates, so the kernel need not relax to this equilibrium from every initial condition. Cosmological recombination requires richer atomic and radiative physics. [3, S3, S4]

Observation and quantum terminology

Quanta move between records named superposed and collapsed when probe coverage or a threshold interaction rule triggers materialization. The current representation does not evolve a general many-body wavefunction or density matrix. Environmental decoherence concerns suppression of interference under interactions; objective-collapse theories introduce specific testable changes to dynamics. A software threshold is not a derivation of either process. In this baseline, collapse is best treated as an effective state transition. [4, 5]

The roadmap's original perception-centric wording is narrowed by BUILD0: perception should govern detail rather than existence. Yet probe-triggered collapse also removes quark rows from weak conversion in covered cells. Existing probe-equivalence checks are limited to selected thermal-population and removal observables. Full observer independence of the baseline is therefore unestablished. Physical measurement interventions, numerical refinement and display visibility need separate contracts. [S2-S5]

Replay and skipping

Random values come from a deterministic counter-based function. EventContext records event keys and payload digests; replay checks event order, periodic state hashes and the final hash. These hashes are not cryptographically linked. The evidence supports same-machine, same-build reproduction. Seeded pseudorandom draws do not establish intrinsic quantum randomness; the implementation does not ingest an external quantum entropy record. Quiet-cell jumps match reference evolution in selected tests. That observed equality does not prove equivalence for arbitrary runs or future coarse-graining. [S4, S5]

5 Observed results and their interpretation

The original audit's seed-42 realization reproduced the archived M4 summary exactly. The final state hash and all 34 hashes matched during replay. Table 2 reports model outputs with enough precision for comparison, while distinguishing acceptance criteria from empirical validation. The experiment is one realization, not an ensemble estimate. [S6, S7]

Observable	Original audit result	Interpretation
Initial settings	8 cells; 3,400 ticks; seed 42	Same configuration as archived M4
Neutron/proton ratio at freeze-out	0.129890	Inside internal band 0.10 to 0.20
Tracked freeze-out temperature	1.08087 MeV	Model-dependent diagnostic
Helium-4 mass fraction Yp	0.227618	Inside internal band 0.20 to 0.30
Hydrogen mass fraction XH	0.747344	Inside internal band 0.69 to 0.80
D/H and Li/H	Both zero	D/H admits zero; Li/H is record-only
Reported ionized fraction Xe	0.00345722	Charge-based electron ledger statistic
Reported neutral fraction	0.996543	Bound-electron fraction 1 - Xe; not neutral-atom number fraction
Nucleons formed; nuclear reactions	1,318; 247	Small finite-population experiment
Jump events; skipped cell ticks	6; 6,000	Cell work omitted, not 6,000 global ticks
Final model time; temperature	4.87921×10^{9} s; 4.52715×10^{-6} MeV	Approximately 155 years under Eq. 1

Table 2. Original audit run values from audit_diagnostics.json and fresh_m4/summary.json. Full precision is preserved in the supplementary data.

Pitrou and colleagues reported $Y_p = 0.24709 \pm 0.00017$ for their precision calculation. The baseline's value is about 7.9% lower in relative terms. This comparison supplies scale, not a statistical significance: the models do not share a calibrated cosmological specification or uncertainty budget. SimWorld's broad internal band is a regression criterion. [1]

Cooke and colleagues measured $D/H = (2.527 \pm 0.030) \times 10^{-5}$ in a seven-system analysis. With only roughly 10^3 hydrogen nuclei, the expected deuterium count at that abundance is about 0.03. A zero count is unsurprising and cannot test that abundance. An ideal Poisson count near 100, needed for roughly 10% counting precision alone, would require about four million hydrogen nuclei. This is a sampling-scale estimate, not a proposed universal particle threshold. [2]

6 Validation gaps exposed by the audit

Passing tests establish the properties they assert. The suite checks replay, tamper detection, selected conservation identities, cache behavior and acceptance bands. The targeted diagnostics expose stronger equilibrium and independence properties that those tests do not establish. Current source comments acknowledge the limitations; some frozen specifications still describe intended guarantees more strongly. The findings below remain unresolved in the executable baseline. [S3-S5, S7, S8]

Finding	Evidence	Consequence
Reverse nuclear pathways depend on forward populations	network._rates skips a pair when either reactant is absent. A product-only diagnostic omitted the p+n reverse channel.	Products cannot dissociate through that channel until the required reactants already exist.
Reverse rates do not implement the stated channel ratio	The reverse expression uses product binding B rather than reaction Q and lacks the forward Gamow factor.	The documented detailed-balance claim needs a derivation and equilibrium tests.
Fully neutral atomic populations are absorbing	100 neutral H nuclei at Tgate/1.2 produced zero events over 1,000 ticks; the internal Saha function gave Xe = 0.10815.	The kernel does not relax from this state to its own nonzero ionization reference.
Random draws reuse stream prefixes	Calls sharing seed, tick, kind, cell and sub restart at the same counter despite different payloads.	Repeated species calls can be correlated; reproducibility does not imply independence.
Storage differs from the performance prototype	Main quant records are dictionaries and lists; Spike A uses array operations.	The prototype throughput cannot be attributed to the current engine.

Table 3. Source findings and targeted diagnostics. The nuclear test allowed other channels to react; the finding is specifically the missing dissociation pathway, not an absence of all reactions.

For closed systems intended to relax to thermal equilibrium, the kernel must admit the correct equilibrium distribution and the intended reverse pathways. Microscopic detailed balance requires $\pi(x)q(x\to y) = \pi(y)q(y\to x)$ at equilibrium π for each reversible transition pair. Open driven systems need not satisfy global detailed balance. A paired rate formula is insufficient if enumeration removes the reverse move or scheduling changes its effective probability. These issues need resolution before the yield bands support physical claims.

The project's own surprises ledger also records imposed baryon excess, flavor bookkeeping approximations, unresolved deuterium and lithium behavior, and recombination much earlier than the physical epoch. Photon and entropy dynamics remain inadequate for a quantitative recombination claim. ETA and EPS_B are declared constants but do not actively control the run; the quark excess is preserved through removal exemptions. Parameters must be traced to operative code before they enter sensitivity studies. [S3, S4]

7 Prototype results and computational feasibility

Two prototype studies inform the roadmap, but their evidence has a different scope from the integrated run. Spike A measures operations on array storage; Spike B tests a fitted approximation to first ionization energies. Both are useful feasibility probes when their workload and calibration are stated. Neither demonstrates a scalable universe simulator. [S6]

Prototype	Recorded result	Limit of inference
A quant storage and update cost	281.4 million quanta/s for an evolve kernel; 20.4 million interactions/s; 13.899 ms mean full tick	One synthetic workload on the recorded host. Main QuantStore has a different representation.
B shell approximation	41 training examples; fitted screening; 6.9% training RMS relative error	Empirical fit, not a self-consistent-field atomic solver or general chemical model.
B selected probe elements	Reported errors: He -16.9%, Li +4.4%, C -18.2%, Fe -4.1%	Small probe set; H uses an analytic anchor. No broad out-of-distribution or molecular validation.

Table 4. Archived prototype measurements from spikes/a_quant_cost/report.json and spikes/b_shells/report.json. No new prototype timing benchmark was performed.

The current main engine allocates and updates Python objects in paths that the original cost model expected to be array operations. A future storage refactor can be justified by a measured bottleneck and checked by replay or appropriately defined statistical equivalence. It must then be benchmarked as an integrated workload including reaction discovery, event logging, hashing, memory traffic and observation changes.

The proposed scale targets include a four-billion-year planetary run with roughly 10,000 resolved elements, a compressed 500-million-year biological era, and 10,000 compressed social agents with at most 1,000 instantiated. These are unmeasured planning targets; future-phase performance tables contain no measurements. Raw quanta per second cannot price chemical discovery, sparse ecological change, long memory or a sudden request for high detail. Each scale needs a workload model with peak memory, log growth and reconstruction cost. [S1: PHASE7, PHASE8 and PHASE11]

Derived caches are compatible with the research objective when their provenance, domain and error are known. Effective multi-level recombination is an example of reducing recurring computation by separating atomic calculations from cosmological evolution. Its success in a defined problem does not license arbitrary compression of chemical or social histories. [3]

A useful runtime report should show a fidelity-versus-cost curve over multiple problem sizes and seeds, plus the worst observed transition into detailed simulation. Hardware, implementation versions, warm-up, repetitions and variability belong alongside throughput. Until those measurements exist, CPU/GPU partitioning and civilization-scale capacity remain design proposals.

8 Architecture for a multiscale research program

The strongest architectural idea in the roadmap is a persistent causal record shared across levels. Bound objects, materials, organisms and institutions should carry enough provenance to explain their creation and subsequent influence. Each transition must define which information is retained, what is approximated and which future queries remain answerable. Figure 1 separates the present reduced model from proposed layers. [S1-S4]

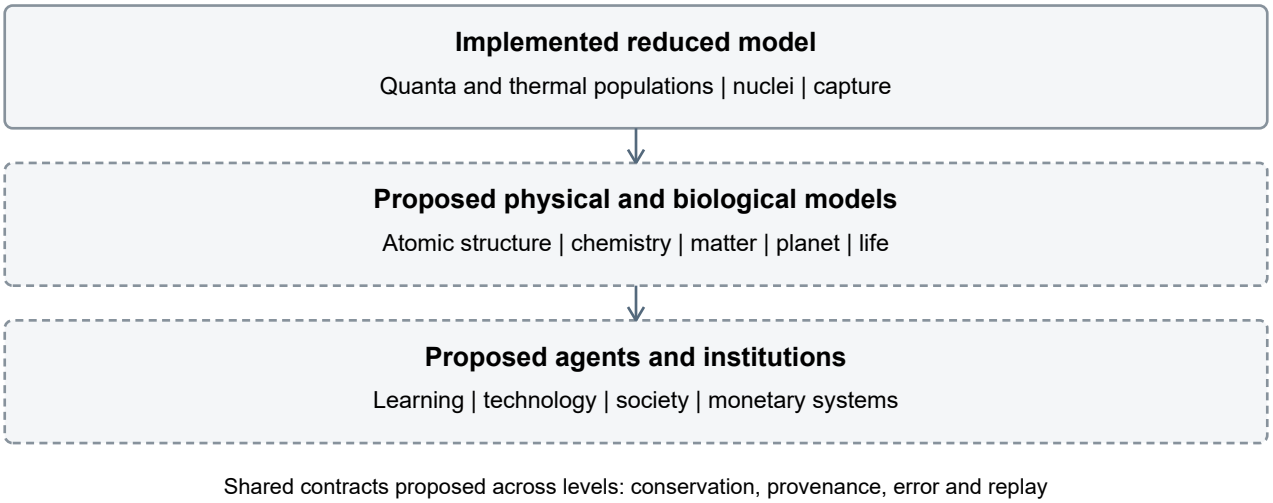


Figure 1. Scope of the current implementation and the proposed hierarchy. Solid outline denotes executable baseline scope; dashed outlines denote future systems. Arrows indicate intended dependencies, not demonstrated end-to-end emergence.

Four contracts are needed at each interface. A **state contract** declares units, conserved quantities, sufficient statistics and boundary inputs. An **event contract** specifies causality, scheduling, random-stream identity and attribution. An **error contract** identifies discarded information and the observable error introduced by compression. A **reconstruction contract** states how detail is regenerated and which histories remain compatible with the summary. These are recommendations that extend the existing interface discipline.

For a fine state x, a restriction R(x) produces a coarse state and a lifting L reconstructs compatible detail. Exact equality L(R(x)) = x is generally impossible after information is discarded. Instead, the model should identify an observable O and assess |E[O(x)] - E[O(L(R(x)))]| against a declared tolerance, while separately testing conservation and distributional differences. Exact replay of the chosen coarse algorithm is a different requirement. [6]

Conservation does not mean that every quantity is closed in every experiment. Expansion, radiation, food, external stress or migration can cross boundaries. The contract must expose those fluxes so a budget closes over state changes plus transfers. Ownership records, knowledge and money add informational claims about physical resources; they must not silently create matter or energy. [S1: PHASE5-PHASE12]

Observation should request refinement only when the retained state can support it. Unseen regions can still affect future events through radiation, transport or agents. Omitting these influences because they are off-screen would change the modeled system. This distinction belongs in the earliest architecture, even though the roadmap assigns perception and runtime their own later phases.

8 Architecture continued

Feedback experiments across levels

The proposed world is more than a sequence of increasingly complicated objects. Later systems must alter the conditions that support them. Three cross-layer experiments illustrate this intended feedback and provide narrower research targets than an entire history. None is implemented in the current micro-seed. [S1: PHASE8-PHASE12]

Ecology and planetary reservoirs. A population consumes substrates, releases products and changes local nutrient and oxygen availability. Transport redistributes those changes, which then alter the population's growth and competitors. Compare a fully coupled system with one in which environmental feedback is held fixed while the same external energy enters both. Measure reservoir closure, population persistence and the response to a resource shock. This tests whether ecological effects are carried through the physical interface rather than assigned as a global vegetation bonus.

Cooking materials and learned behavior. An agent changes a material through heating, and the resulting chemistry changes usable energy or tool performance. Those returns influence future choices, while fuel consumption changes the environment. A controlled experiment can hold the sensory and learning model fixed and vary access to the physical transformation. Observable outcomes include net energy after fuel and processing costs, transfer of a learned procedure to unfamiliar resources, and persistence of that procedure across agents. A successful result would support one coupled mechanism; it would not establish an inevitable route to civilization.

Industry exposure and collective maintenance. Production moves material and releases pollutants; transport determines exposure; agents perceive consequences and decide whether to maintain, regulate or abandon an asset. Compare matched communities under different communication reliability and maintenance costs. Measure exposure, resource use, repair effort and the distribution of outcomes. Ownership and obligation records can attribute choices and costs, while the physical model determines where the pollutants go. Political labels should be assigned only after the behavior is characterized.

What compressed histories must retain

The biological plans retain genotype frequencies and host-parasite correlations. The cognitive and social plans add skill and role distributions, ancestry and correlations between lineages and culture. These joint relationships matter: independently preserving a skill histogram and a resource histogram can lose which agents can use which resources. Stable periods may permit large temporal jumps, while shocks or strong interactions require detailed evolution. [S1: PHASE8-PHASE11]

Tests should reconstruct detailed populations from a summary and compare response distributions to matched shocks. The closure is inadequate if identical coarse records hide histories that predict materially different futures for the chosen observables. Retaining more correlations may improve fidelity at a memory and computational cost. Measuring that tradeoff is a central contribution the project could make, even if the full end-to-end history remains out of reach.

9 Roadmap from world rules to continuum physics

Table 5 translates phases 0 through 6 into research deliverables, combining selected source gates with the validation requirements developed in this paper. They are proposed requirements unless identified as demonstrated earlier. The micro-seed spans simplified nuclear and atomic behavior within the Phase 0 build; this does not mean that the full Phase 2 or Phase 3 programs are complete. [S1, S2]

Phase and intended mechanism	Required evidence	Present status and main risk
0 World rules Quant records, fields, time, observation and replay	Frozen state/event contracts; runnable cooling seed; measured cost and replay	Reduced baseline exists. Physical interpretations exceed what representation tests establish.
1 Quantum layer Array storage, field kernels, interference, entanglement and scheduling	Interference controls; non-signalling statistics; spectra; reproducibility and throughput	Detailed proposal. Correlated draws or threshold collapse alone are insufficient.
2 Nuclear layer Residual forces, binding, reactions, decay and energy deposition	Held-out binding curves; light nuclei; stellar yields; event conservation; decay closure	Detailed proposal beyond toy kernel. Fitted anchors must be separated from predictions.
3 Atomic layer Radial self-consistent fields and correction hierarchy	Convergence; ionization and affinity; spectra; high-Z limits; molecular suitability	Shell-fit spike exists; proposed solver is unbuilt. Atomic trends do not establish chemical accuracy.
4 Chemistry Bond curves, geometry, channel search, barriers and surfaces	Molecular energetics; kinetics; detailed balance; polymer and reactor controls	Detailed proposal. Local search cannot guarantee discovery of every relevant pathway.
5 Materials Composition, microstructure and processing history	Mechanical and transport curves; hysteresis; phase behavior; reconstruction errors	Detailed proposal. Bounded history may omit decisive path dependence.
6 Continuum physics Gravity, fluids, heat, stress, radiation and electromagnetism	Convergence and conservation; orbit, convection, fracture, wave and engine benchmarks	Detailed proposal. Constitutive closure and coupling across scales remain untested.

Table 5. Physical phase map from PHASE0.md through PHASE6.md with proposed validation refinements. Criteria are research targets, not reported completions.

The phase order should be interpreted as a dependency graph rather than an obligation to finish an entire discipline before testing a connection. For example, a small thermally driven reaction system can test chemistry, heat exchange and logging before any planetary model exists. Such demonstrations reveal incompatible units, missing reverse processes and closure failures earlier than a full-world build.

The phrase every combination is calculable needs an operational domain: allowed species, conditions, approximation level, search budget and failure behavior. A simulator should report an unsupported state or unresolved pathway explicitly. Quietly returning no reaction would confuse model incompleteness with physical impossibility.

10 Roadmap from planets to institutions

The middle phases add environments, reproducing systems and agents that act through the physical world. Their common requirement is that higher-level summaries remain connected to resource flows and observable events. Their greater complexity also increases the risk of calibrating a preferred narrative instead of testing a mechanism. [S1]

Phase and intended mechanism	Required evidence	Primary research uncertainty
7 Planet and space Accretion, differentiation, climate, reservoirs and stellar summaries	Accretion/differentiation; closed water and carbon budgets; climate and LOD comparisons	Geological closure and long-time drift; seeded prior stellar output must be disclosed.
8 Life Templating, compartments, metabolism, heredity and ecosystems	Catalytic closure; persistence; heritable variation; competition; energetic and population agreement	Abiogenesis and complex bodies are open outcomes. Seeded replicators change the claim.
9 Agency and civilization Learned policies, bodies, tools, cooking and culture	Novel-resource tasks; tool performance; energy viability; cumulative learning; village comparisons	Long discovery chains and reward design may select the desired story.
10 Technology Experimentation, instruments, machinery, communication and computation	Energy/entropy closure; signal and grid tests; reproducible invention; nested replay	Hardware emergence and host-side nested simulation need an explicit equivalence test.
11 Society Communication, sanctions, learning, coalitions and institutions	Matched interventions; language/knowledge patterns; exposure-response; compression comparisons	Many mechanisms fit similar patterns. Human social categories need operational definitions.
12 Money and politics Ownership, obligations, credit, enforcement and diffusion	Consistent holdings; defaults and bank runs; real/nominal separation; conditional regime tests	Institutional outcomes are contingent. Accounting consistency does not validate behavior.

Table 6. Proposed phases 7 through 12. No implementation of these phases was identified in the inspected sim package.

Phase 7 permits disk composition seeded from an offline stellar calculation, and Phase 8 discusses a disclosed seeded-replicator fallback. Both can support useful experiments. They must be labeled as initialized boundary conditions rather than evidence that the entire preceding history emerged in the same realization. A continuous data record cannot by itself supply the missing causal history.

Phases 9 through 12 should evaluate distributions of possible histories, conditional on environments and agent rules. Fire, farming, computers, currencies or a particular political regime cannot be both guaranteed milestones and independent emergent predictions. Failed discovery, extinction and alternative institutions must remain admissible outcomes. The test protocol should specify what each outcome teaches before the seed is run.

11 Integration and unresolved dependencies

Phase	Detailed proposed mechanism	Proposed evidence
13 Infrastructure	Owned networks, maintenance histories, producer-attributed emissions and local exposure tracers	Transport and attribution closure, maintenance-dependent failure, repair and recycling with physical delays.
14 Perception	Agent-derived gaze, memory/testimony provenance, deterministic frame allocation and detail governor	Equivalent behavior under matched maps, no privileged sensing, replayable allocations and bounded epistemic differences.
15 Runtime	Data-oriented execution, CPU/GPU scheduling, build fingerprints and certified coarse intervals	Same-build exact replay, declared cross-build error bounds, measured budgets and refusal of uncertified summaries.
16 Integrated loop	Global event ontology, frozen read-only milestone detectors, multi-seed runs and in-world nested simulation	Recorded ladder and network-mesh evidence, negative controls, varied histories, replay, playability and cost reports.

Table 7. Phases 13 through 16 now have detailed specifications. Their mechanisms and gates are proposed, not implemented results. Appendix B provides the complete phase profiles.

The current seams ledger contains 25 entries: 22 OPEN and three FIXED wording or interface-contract entries, S14, S17 and S21. The latter statuses do not demonstrate implementation of the associated downstream mechanisms. New dependencies S19-S25 address emissions attribution, local exposure transport, agent-derived gaze, deterministic frame allocation, runtime fingerprints, certified jump intervals and a global event ontology. Several downstream plans require these upstream changes before their own gates can be evaluated. [S3, S8]

Perception and runtime remain cross-cutting dependencies even though they now have detailed Phase 14 and 15 plans. Observation maps must derive from simulated agents; frame budgets must depend on simulated complexity rather than host load. Coarse intervals require certificates supplied by the owning model, and runtime-build fingerprints distinguish exact same-build replay from bounded cross-build comparisons. A scheduler cannot safely invent semantics that an upstream model has not defined. [S1: PHASE14-PHASE15]

Phase 16 adds a global event ontology and read-only milestone detectors, an internet-like mesh-growth milestone, and nested simulation triggered by in-world actions. Proposed multi-seed runs and negative controls can test whether declared detectors and input restrictions behave as intended. They cannot prove universal emergence, guarantee a particular history, or establish equivalence between a host-side child engine and physical in-world computation. The prescribed ladder remains a program acceptance target whose scientific interpretation requires independent controls. [S1: PHASE16]

A phase completion record should identify the source revision, calibration dataset, held-out tests, known failure domain, measured costs and the next interface demonstrated. Checkmarks alone are insufficient. This record would make the roadmap a cumulative research program in which a failed validation can revise the model without requiring the entire project narrative to remain intact.

12 Experimental design and uncertainty

Verification before empirical comparison

The next experiments should establish that each stochastic kernel samples the model it claims to sample. Test reversible systems from both sides of equilibrium, including zero-reactant and fully occupied states. Compare small reaction networks against an analytic stationary distribution or a trusted stochastic reference. Separate errors in reaction enumeration, rates and event sampling. A time-step refinement study is meaningful only after physical time and per-step rates have a consistent relationship. [8]

Conservation tests should account for all modeled reservoirs and boundary fluxes, rather than only algebraic identities derived from the same state variable. Tests should also expose forbidden transitions, negative populations, inconsistent unit conversions and changes caused by event ordering. Replay corruption and truncation tests remain valuable because they establish integrity of the experimental record. [S5]

Calibration and independent validation

Before adjusting a parameter, declare which observables calibrate it, which are held out and which conditions lie outside the model's domain. Nuclear anchors and freeze-out/yield bands used to tune constants cannot simultaneously serve as independent evidence for those constants. Held-out isotope systems, temperature regimes, compositions and initial populations are more informative than rerunning the fitted cases.

A proposed initial study would use fixed calibration data, at least 30 held-out seeds, and additional seeds until uncertainty intervals stabilize. Thirty is a planning starting point, not a guarantee of power. Held-out seeds assess stochastic robustness; independent empirical validation still requires external observations or trusted reference solutions. Rare-event designs depend on expected counts and confidence limits. Stream overlap must be corrected or accounted for before independent-sample calculations are applied.

Propagation across scales

Uncertainty includes stochastic sampling, parameter uncertainty, numerical discretization, missing mechanisms and model reduction. These components should be reported separately. A compact organizing relation is $y = F(\theta, s, h, C)$, where θ denotes parameters, s the random seed, h the numerical resolution and C the chosen closure. Varying only s measures neither parameter sensitivity nor structural error.

For each compression level, compare detailed and reduced runs under matched conditions. Report bias in primary observables, distributional distances, conservation residuals and rare-event rates as functions of computational cost. Correlated errors can accumulate across interfaces, so validating each layer in isolation does not bound the error of the combined world. Cross-layer interventions are needed to identify where a result becomes sensitive to a closure. [6]

Publish negative results and the settings that produce them. A model of emergence is especially vulnerable to selecting successful seeds or adjusting rules until a desired transition appears. Predeclared observables and all-seed reporting make it possible to distinguish a reproducible mechanism from a curated history.

13 Priority experiments and decision criteria

The following sequence is proposed work. It is ordered to remove uncertainties that would otherwise propagate into every later phase. Completion requires the associated evidence, not simply an implementation or a visually plausible trajectory.

Priority experiment	Protocol and primary observable	Criterion for advancing
1 Reversible reaction kernels	Product-only nuclear initial states; neutral and ionized atom states; multiple fixed temperatures; reference stationary distributions	All intended reverse pathways exist; the same equilibrium is reached from both sides within predeclared statistical error.
2 Random-stream identity	Distinct species/events, reordered independent work and parallel scheduling; audit counter uniqueness and correlation	Unique documented keys for independent draws; stable replay; passing independence and scheduling checks.
3 Frozen physical baseline	Dimensional rate mapping; conservation budgets; resolution and step refinement; held-out nuclei and atomic cases	A published error budget and domain of validity; calibration and prediction results reported separately.
4 Measured runtime model	Integrated engine at several population sizes and observation patterns; repeated timings and memory/log accounting	A defensible cost curve for the actual implementation at the required fidelity.
5 One coupled physical experiment	A small reaction system exchanging heat with a reservoir, with detailed and reduced representations	Mass/energy budgets close; kinetic and thermal observables agree with reference within declared tolerances.
6 Open-ended extension	Add compartments after validated chemistry, or tool use after validated mechanics, sensing and learning; test controls and new environments	The measured new behavior depends on the proposed mechanism and survives held-out conditions.

Table 8. Recommended experimental order. Numerical tolerances must be chosen for each observable and benchmark before evaluating the results.

Every experiment should archive its model version, parameter values, inputs, initial state, random-stream schema, raw outputs, analysis procedure and failure cases. Changes to physics or random-stream identity require a new baseline. Historic files should remain replayable with their original code or be explicitly marked as requiring an archived environment.

The first coupled experiment is deliberately much narrower than a planet. It tests the project's central proposition: that independently checked mechanisms can be composed without losing accounting or explanatory value. Success would justify a larger coupled system. Failure would locate an interface problem before it is hidden by biological or social complexity.

14 Research questions for life and social emergence

From reaction networks to reproducing systems

An autocatalytic network is an intermediate chemical property. It does not by itself demonstrate heredity, selectable variation, compartment persistence or open-ended evolution. Research on autocatalytic sets explicitly separates catalytic closure from a complete account of life's origin. Phase 8 should therefore report these properties individually and use non-catalytic, non-templating and resource-limited controls. [9]

A falsifiable question is whether templating increases lineage persistence and heritable variation under a fixed resource flux compared with a matched network without templating. A second is whether compartments change competition or extinction rates after controlling for total material and energy. These comparisons need direct chemical and population observables, not merely a detector that labels a pattern as life.

Digital evolution has shown complex functions emerging in engineered populations of replicating programs. It supports controlled studies of novelty and historical contingency, while supplying no proof that a physical chemistry model will generate life or human-like intelligence. Any seeded replicator or designed reward must be declared as part of the experiment. [10]

From agents to knowledge and institutions

For agency, separate what an agent can sense, remember, infer and physically do. Tool-use success should depend on the modeled mechanics and resource cost. Research claims should compare learners with restricted memory, communication or experimentation and quantify task performance, energy expenditure and transfer to unfamiliar environments. A novelty measure must distinguish new behavior from relabeling an existing action.

For cumulative knowledge, test whether communication and retained records improve performance across generations under matched ecological conditions. For economic organization, test whether alternative ownership, credit and enforcement rules change exchange, defaults, production and inequality. For institutions, use interventions on mechanisms and report the range of outcomes across seeds. ODD-style documentation can expose scheduling and initialization choices that otherwise resemble emergent social laws. [7]

One proposed institutional experiment compares otherwise matched resource-sharing communities with different communication reliability and sanction costs. Resource persistence, enforcement expenditure and outcome distributions would distinguish successful coordination from a prescribed institutional label. Initial endowments, agent capabilities and external shocks must be matched, and failures retained.

The proposed social variables are operational abstractions. They do not yet justify claims that mental health follows directly from a small neurochemical state, that a named culture has a fixed mechanism, or that a political system has a universal performance ordering. Human-facing interpretations require independent empirical work and clear limits on what the agents represent. The value of the simulator at this stage would be testing consequences of explicit assumptions, not assigning scientific inevitability to a preferred society.

15 Discussion and research contribution

SimWorld's roadmap is most useful as a program for testing interfaces between effective models. Its breadth creates a demanding requirement: every upward summary must preserve the consequences that the next level relies on, while every downward action must be paid for in the relevant physical or informational budget. A coherent history is valuable only when the meanings of its events remain stable across those transformations.

The existing build executes a cooling sequence, produces aggregate bound states, records stochastic causes and reproduces its output. Windows and Linux CI now verify the regression suite independently, and the repository publishes a static research library. Explanatory comments have improved, but the executable model remains unchanged: structural stability filters shape outcomes, reverse reactions can disappear from enumeration, and a neutral state can remain fixed when the internal equilibrium helper predicts ionization. [S4, S8]

These limits do not make the reduced model uninformative. They define which experiments it can currently support and which claims must wait. The immediate scientific opportunity is to use its reproducibility machinery to isolate and correct kinetic assumptions, establish independent benchmarks and quantify the consequences of approximation. Those steps are more informative than extending the timeline before the basic kernels have a validated domain.

The project should also distinguish three senses of emergence. **Computational emergence** is a pattern generated by specified local rules. **Robust emergence** persists across reasonable parameter, resolution and initialization changes. **Empirical adequacy** is agreement with the intended real system on independent evidence. A phenomenon can satisfy the first definition without the second or third. This distinction makes success measurable while preserving room for useful exploratory worlds.

The long-term claim of an unscripted route from a quantum seed to civilization remains open. The phase documents already introduce effective fields, fitted forces, reduced chemistry, constitutive material rules, learning mechanisms and seeded boundary conditions. These are legitimate modeling tools when disclosed and tested. The program should measure what causal information they preserve rather than describe the whole stack as a deduction from a minimal set of universal constants.

Conclusion

The current evidence supports a reproducible reduced simulation and a detailed agenda for its extension. It does not yet support quantitative cosmological fidelity or completed higher-level emergence. Progress toward a research-grade platform should be judged by reversible and benchmarked kernels, independent validation, explicit cross-scale error contracts and reproducible experiments that can fail. A small validated chain of coupled mechanisms would be a stronger next contribution than an unvalidated realization of the entire roadmap.

Appendix A Reproduction and data provenance

The research archive separates source/ at commit 4134abd from baseline_source/, the original audit snapshot with public identifier redactions. It preserves the original run and diagnostic files, and adds the v1.1 manuscript, all 17 profiles, publication status and separate SHA-256 manifests. The tracked research/whitepaper directory in the repository contains the editable publication sources. No new physical experiment is claimed by this document revision. [S7, S8]

Environment and commands

Engine verification used Python 3.11.9, NumPy 2.4.6 and pytest 9.1.1. The platform API reported Windows-10-10.0.26300-SP0; the build specification calls the machine Windows 11. These labels are recorded without inferring operating-system identity from the API string alone. Execute the following from the source directory with the stated Python environment:

```
python -m pip install -r requirements.txt
python -m pytest -q
python -m sim.demo --seed 42 --ticks 3400 --cells 8
  --probe-cells 2,3 --out reproduced_m4
python -m sim.replay reproduced_m4
```

The two displayed demo lines are one shell command; join them with a space. Installing dependencies changes the selected Python environment, so an isolated environment is appropriate for reproduction.

Identity checks

The original audit summary matched baseline_source/_m4run/summary.json exactly. Both hash sequences matched. The following fingerprints permit exact comparison without treating rounded paper values as the reproducibility target:

Final state SHA-256
`fe8e094a9753b27c89fc0dffd1905f8de4be`
`45263158d5712618ddd5f6ded1a0`

Decompressed event-log SHA-256
`15e50cd6adbc6cc0f01360ebd6fc4a8a97a7`
`06e0cb459f0fd218abeafba777f3`

Each displayed fingerprint is one 64-character hexadecimal string; remove the visual line break.

The original audit run took 5.550 seconds and its replay 5.547 seconds while pytest was also active. These observations under concurrent load have no repeated-trial confidence interval. The full-suite JUnit report and text output preserve the 65-pass result. The older nightly report is historical evidence only. To repeat the diagnostic cases from the archive root, use:

```
python research/audit_diagnostics.py --source-root baseline_source
  --output-dir reproduced_audit
```

Join the two displayed lines into one command. The output directory must be new, preserving archived evidence.

Scope of reproducibility

Exact replay equality is supported within the audited source and environment. Independent Windows/Linux CI success does not itself compare complete hash sequences across platforms. The archive contains separate current and baseline source manifests and the status record identifies the exact tested commit and workflow runs. The source baseline is fixed before this publication commit, preventing a circular claim that the PDF embeds its own containing commit hash. [S8]

Appendix B Phase 0 World rules

Status A reduced Phase 0 prototype is implemented. Several intended guarantees remain unestablished; later phase completion must not be inferred from this baseline.

Phase 0 defines the state, inputs, observation semantics and time model that later layers must inherit. Its central deliverable is a runnable physical seed with an inspectable history, establishing an engineering baseline before higher-level emergence is attempted.

Proposed mechanisms

The plan organizes this work into five decisions: quant representation, effective fields, initial conditions, observation and temporal advancement. BUILD0 adopts a smaller effective field set and a geometric cooling schedule for the micro-seed. Current quant records are dictionaries and lists rather than the proposed packed array store. Effective binding and aggregate electron capture produce a small stochastic history; event keys and periodic state hashes allow re-execution checks. Quiet-cell jumps omit selected work under restricted eligibility conditions. Separate storage-cost and shell-approximation spikes inform later development without becoming measurements of the integrated engine.

Prerequisites and interfaces

There is no earlier phase dependency. The phase must publish versioned constants, state invariants and read/write boundaries. Its original choices also constrain later perception, gravity amendments, runtime-build fingerprints and nested instances; those future requirements should be tracked as explicit interface revisions rather than silently added to the original baseline.

Proposed validation

The proposed gates include record round-trips, measured storage/update costs, observable probe behavior, light-element output checks and replay equivalence. The actual audit supports 65 passing regression tests and a reproduced seed-42 history with 34 matched hashes in the tested environment. These checks do not validate general quantum dynamics or precision cosmology. Further work should test equilibrium from contrasting initial states, random-stream separation and conservation across every conversion boundary, while keeping fitted targets separate from independent physical benchmarks.

Research limits

The largest risk is treating a coherent implementation as a physical derivation. The observation threshold is an effective software rule, fitted binding is not a fundamental force calculation, and the array microbenchmark does not describe current dictionary storage. Exact replay also establishes neither statistical independence nor a complete resumable checkpoint.

Sources: PHASE0.md: The five decisions (task list); PHASE0.md: How to implement (sequencing); PHASE0.md: Risks; PHASE0.md: Carry-over open questions (from ROADMAP); BUILD0.md: Decisions; Engine architecture; spec/seams.md: S4, S5, S9, S16, S22, S23. [S1, S3]

Phase 1 Quantum layer

Status Detailed proposal. The baseline provides precursor logging, sampling and probe behavior, but the full array store, interference dynamics and correlation validation are not implemented.

Phase 1 proposes the reusable microscopic execution layer: storage, field interactions, observation, correlations and scheduling. It would replace the micro-seed's provisional structures with interfaces that later nuclear models can consume without depending on storage internals.

Proposed mechanisms

The planned store uses array blocks indexed by cell and refinement tier, with one stable identifier space across superposed and collapsed records. An explicit edge table represents correlations, and counter-based streams address stochastic events. Field kernels sample interactions from an effective coupling table. A tier scheduler promotes detailed regions and propagates quiet summaries containing temperature, composition, free energy and entropy obligations. Collapse is a logged threshold or coverage operation. A scripted probe remains a test fixture; Phase 14 is intended to supply the world-internal observation map through a compatible interface.

Prerequisites and interfaces

Phase 0 decisions, constants and boundary contracts must be closed with evidence. The replay harness becomes continuing infrastructure, while measured spike results inform, rather than substitute for, integrated budgets. The nuclear boundary allows reads of collapsed records and composition summaries, and writes of binding outcomes through the field-interaction contract. S21 clarifies the future observation-map producer.

Proposed validation

Proposed tests include pack/unpack invariants, storage costs at large synthetic populations, interference controls, non-signalling correlation statistics, cavity spectra and repeated deterministic re-execution. The plan targets at least ten million superposed updates per second, but its future performance table remains unmeasured. Skipped and fully evolved regions require both an approximation comparison and replay of the selected algorithm. The acceptance specification also retains a light-element integration test, although another passage excludes nuclei from the standalone layer; that test boundary needs explicit reconciliation.

Research limits

Naming a field, phase variable or entanglement edge does not establish a valid quantum model. Observable interference, marginal distributions and measurement interventions need an explicit mathematical specification. Log growth, correlation-graph complexity and ordering-dependent random keys can undermine scalability or independence even when a single execution replays exactly.

Sources: PHASE1.md: Architecture; PHASE1.md: Work packages W1-W6; PHASE1.md: Performance budget (fill from Spike A, then hold the line); PHASE1.md: Interface exposed to Phase 2 (nuclear); PHASE1.md: Verification strategy; spec/seams.md: S21, S22, S23. [S1, S3]

Phase 2 Nuclear layer

Status Detailed proposal. Current fitted binding and bounded reaction counts are demonstration kernels, not the force-based nuclear program described here.

Phase 2 seeks a nuclear model whose bound states, reaction pathways and radiation outputs are computed from declared effective interactions. It would supply atomic structure with consistent nuclide records and give later environments a traceable source of composition and ionization energy.

Proposed mechanisms

Nucleons are proposed as composite valence states; nuclei retain constituent references, excitation energy and derived scalar properties. A residual attractive interaction with short-range repulsion and saturation combines with the existing electromagnetic coupling. Barrier sampling supplies fusion probabilities, while stability gradients and penetration models govern candidate fission and decay events. Radiation products reuse lower-layer species and deposit energy into cells. A versioned nuclide registry memoizes computed quantities but must remain regenerable. Event scheduling concentrates work on changes in temperature or composition instead of scanning every nucleus at every tick.

Prerequisites and interfaces

The plan requires the full Phase 1 acceptance and performance gates, not merely today's micro-seed. Phase 3 receives mass, charge, binding, lifetime and decay information through nuclear::nuclide; electron capture may request a transmutation that the nuclear layer adjudicates. Later planetary initialization also depends on a separately logged stellar-yield calculation under the proposed D6 decision.

Proposed validation

The binding model would be calibrated on a small declared anchor set and evaluated on roughly 200 other nuclides. Tests cover light-nucleus structure, the binding-energy maximum, solar-like burning, carbon production at hotter conditions, terminating decay chains and radiation-energy accounting. Random initial states and a million logged transitions should audit baryon number, charge and energy. Cache deletion must preserve recomputed results. Reaction sampling and hot-region compression also need convergence and reference-distribution checks, particularly after the baseline's reverse-path limitations are repaired.

Research limits

Nuclear rates and half-lives are highly sensitive to structure and barriers. Broad agreement or correct qualitative trends would not establish precise yields or predictive decay times. A proposed hundred-fold lifetime tolerance is an exploratory target. Exact equality after reacting-region compression requires a restricted proof or observable error criterion, not an unrestricted assertion.

Sources: PHASE2.md: Architecture; PHASE2.md: Work packages W1-W6; PHASE2.md: Interface exposed to Phase 3 (atomic); PHASE2.md: Verification strategy; PHASE2.md: Risks; spec/seams.md: S6. [S1, S3]

Phase 3 Atomic layer

Status Detailed proposal. The current capture ledger and separate fitted screening study do not implement the planned self-consistent-field solver or full periodic-table validation.

Phase 3 proposes a computed atomic interface for chemistry and radiation. Its outputs would include orbital energies, densities, ionization and affinity information, characteristic sizes and spectra, with failures and approximation levels recorded explicitly rather than hidden behind element-specific tables.

Proposed mechanisms

A radial self-consistent-field solver on a logarithmic grid starts from hydrogenic states and uses a spherical, frozen-nucleus approximation. A declared correction ladder adds correlation terms and configuration mixing when numerical diagnostics indicate inadequacy. Derived records are keyed by physics build, nuclear charge and ionization state. Energetic differences and density integrals supply electronic properties; level differences and overlap estimates supply spectral data. Occupation-balance and pressure-ionization models connect those records to temperature and density. Lazy evaluation supports unusual species, while planned precomputation covers the standard charge range before interactive use.

Prerequisites and interfaces

The nuclear layer must provide validated charge, mass, radius and binding records through its frozen interface. The shell-fit spike supplies only an exploratory cost/error reference. Chemistry may read atomic records but cannot modify them directly; ionization changes pass through the occupation model, and transmutation remains under nuclear ownership. Spectral consumers share a separate read contract.

Proposed validation

The proposal combines grid-refinement convergence with ionization-energy, electron-affinity, periodicity, spectral-line and plasma benchmarks. It envisages a nightly charge sweep through 118 and controlled comparisons for neutral and ionized states. Cache-off reconstruction, charge identities and energy accounting accompany physical tests. Validation must distinguish valence-electron count from bonding valency, which one source example conflates. High-charge stability claims require a justified relativistic and nuclear domain; the paper does not assume that a neutral atomic table must end at a particular charge.

Research limits

Spherical approximations and simple correlation corrections can fail for near-degenerate shells, anions and molecularly important properties. Matching selected ionization energies does not establish accurate wavefunctions, affinities or bond energetics. Downstream tolerances must determine acceptable atomic errors, with unresolved cases exposed as structured outcomes instead of silently broadened bands.

Sources: PHASE3.md: Architecture; PHASE3.md: Work packages W1-W5; PHASE3.md: Interface exposed to Phase 4 (chemistry); PHASE3.md: Verification strategy; PHASE3.md: Risks. [S1, S3]

Phase 4 Chemistry layer

Status Detailed proposal. No general molecular solver, reaction-discovery engine or chemical surface model is present in the current simulation package.

Phase 4 would connect atomic calculations to molecular structure, reactions, thermodynamics and surfaces. Its research goal is extensible chemistry in which newly encountered species can be represented and evaluated within a declared approximation domain, without requiring a pre-enumerated product catalog.

Proposed mechanisms

The proposal combines diatomic electronic approximations with angular terms for polyatomic geometry. Canonical bond graphs identify species and cache derived formation, dipole and intermolecular quantities. Local valence-compatible graph rearrangements generate candidate products; approximate barrier relations and collision conditions produce event propensities. Surface-adsorbed intermediates provide candidate catalytic pathways. Thermodynamic records connect reaction energies, entropy approximations and intermolecular terms to phase behavior and solubility. All caches are computations that can be repeated, rather than independent authorities. Later seams add explicit channel classes for templating, light-driven reactions and amphiphile assembly that ordinary local rearrangement does not supply.

Prerequisites and interfaces

Validated atomic records must expose the energies, occupancies and spatial information the bonding approximation requires. Materials consume chemical graphs, phase properties, reactions and surface states through chem::species and chem::react. S10-S12 are upstream prerequisites for heredity, compartments and photochemistry; their absence cannot be repaired by downstream biological bookkeeping alone.

Proposed validation

Selected source gates include molecular geometry and energetics, ionic and metallic limits, water's density anomaly, catalytic controls, polymer distributions and an organic reactor. A broader protocol needs separate tests of channel discovery, kinetics and equilibrium. Closed reversible systems should approach the stated equilibrium from contrasting initial states. Atom and energy budgets, valence validity, unfamiliar-species cases and cache deletion are additional checks. A small calibration set, including any catalytic anchor, must be distinguished from held-out molecules, temperatures and reaction classes.

Research limits

A bounded local search is not a completeness proof for chemistry, and internal thermodynamic consistency cannot identify a missing pathway. Pair-and-angle energies and approximate barriers may miss collective electronic effects or rearrangements. A product with a drug-like formation energy does not establish pharmacological function. Search-budget exhaustion must be distinguishable from physical impossibility.

Sources: PHASE4.md: Architecture; PHASE4.md: Work packages W1-W6; PHASE4.md: Interface exposed to Phase 5 (materials); PHASE4.md: Verification strategy; PHASE4.md: Risks; spec/seams.md: S10, S11, S12. [S1, S3]

Phase 5 Materials and substances

Status Detailed proposal. Current engine counts do not implement material microstructure, constitutive properties, processing histories or the proposed mechanics interface.

Phase 5 would summarize chemical populations as bulk materials whose behavior depends on composition, microstructure and processing history. It supplies the constitutive information needed by mechanics while preserving a route back to the lower-scale assumptions and the events that formed each sample.

Proposed mechanisms

Material records combine phase fractions, composition signatures, grain structure, porosity, texture, fiber directions and selected history features. Equilibrium and nucleation/growth models connect cooling and processing to microstructure. Effective mechanical relations derive stiffness, strength and fracture behavior; transport models address conductivity, viscosity, diffusion and permeability. Pressure-temperature-time records support geological materials. History remains bounded by folding older events into selected statistics. The mechanics layer writes deformation or exposure events through mat::note_deformation, after which properties are recomputed; it must not directly edit a material's strength or identity.

Prerequisites and interfaces

The phase requires chemical species, reactions and thermodynamic interfaces, together with explicit inherited error bands. Phase 6 consumes mat::record and the event write-back contract. S3 adds vapor-to-droplet nucleation needed for clouds, S11 adds compartment-assembly kinetics, and S24 requires interval-certified jump summaries for materials before a later scheduler can safely skip their evolution.

Proposed validation

The proposed forge, kiln, press and loom manifest compares approximately 100 material examples, accompanied by thermal and strain-cycle fuzzing. Tests cover cooling-rate effects, work hardening and annealing, phase mixtures, anisotropy, fracture and transport trends. Bounded-history assertions and cache-off replay check representation discipline. Detailed particle-level reference cases should quantify observable errors introduced by summaries, including hysteresis and response to changed loading. The plan's tight trend bands and looser absolute bands are exploratory criteria; both require independent reference data and declared calibration inputs.

Research limits

A short history digest may omit path dependence that controls a later failure or phase transition. Small particle samples may also be insufficient references for bulk constitutive behavior. Materials cannot be treated as the only lossy stage when later planetary and population models also compress state. A conductivity trend alone is inadequate validation for working semiconductor devices.

Sources: PHASE5.md: Architecture; PHASE5.md: Work packages W1-W6; PHASE5.md: Interface exposed to Phase 6 (physics/mechanics); PHASE5.md: Verification strategy; PHASE5.md: Risks; spec/seams.md: S3, S11, S24. [S1, S3]

Phase 6 Continuum physics

Status Detailed proposal. The current shared-temperature lattice provides neither spatial continuum evolution nor the gravity, solid, radiation and electromagnetic operators described here.

Phase 6 would evolve spatial fields of matter, momentum and energy using the material properties derived below it. It connects microscopic and constitutive descriptions to fluids, heat transfer, fracture, radiation and electromagnetic effects that later planets and machines need.

Proposed mechanisms

The plan uses conservative cell fields with advection, diffusion and coupling steps, constrained by local stability limits. Gravity advances orbits and feeds potential-energy changes into budgets. Stress and contact operators produce logged deformation and fracture events. Heat transport and phase change reconcile an entropy ledger with lower-layer thermodynamics. A small set of photon-energy groups uses spectral and material opacity records, while acoustic behavior is read from the continuum operator. Charge/current continuity, induction and magnetic transport complete the initial electromagnetic proposal. Materials are changed through declared event interfaces, not by overwriting their derived properties.

Prerequisites and interfaces

Material records, spectral data, chemical thermodynamics and jump-state accounting must be validated first. Later consumers expose missing requirements: S1-S4 cover rotation, viscoelasticity, cloud nucleation and gravity/versioning; S8 requires logged boundary-condition events; S13 adds articulated mechanics; S15 extends electromagnetic regimes for circuits and radio. These are required research interfaces, not existing capabilities.

Proposed validation

The proposed manifest combines orbit and conservation tests, convection and rain-in-a-box, conduction, a heated piston, contact/fracture, spectral absorption, magnetic amplification and wave propagation. Random field initializations exercise stability limits and numerical pathologies. Cross-coupled work, heat, radiation and boundary fluxes must close their budgets. Mesh, step and operator-order refinement should establish convergence independently of pattern matching. Compression comparisons should report observable errors and ledger residuals; bit-identical re-execution of a reduced path is separate from its agreement with detailed evolution.

Research limits

Coarse transport and sub-grid closures may reproduce a trend while missing its magnitude or mechanism. A magnetostatic component does not validate dynamical circuit or radio behavior. Prescribed future performance budgets remain unmeasured. Conservation identities constrain implementation but do not independently establish the correctness of constitutive laws, radiation transport or turbulent closure.

Sources: PHASE6.md: Architecture; PHASE6.md: Work packages W1-W6; PHASE6.md: Interface exposed to Phase 7 (planet & space); PHASE6.md: Verification strategy; PHASE6.md: Risks; spec/seams.md: S1-S4, S8, S13, S15. [S1, S3]

Phase 7 Planet and space

Status Detailed proposal. No planetary, astronomical, reservoir or spatial climate implementation exists in the current micro-seed; planning commitments remain prospective.

Phase 7 proposes planetary environments and astronomical context over geological time. It would connect accretion, interior evolution, climate and chemical reservoirs through the same physical operators and ledgers, supplying life with environments whose resource and energy sources remain traceable.

Proposed mechanisms

A bodies layer evolves disk material and merger events; planetary summaries retain mass, spin, orbits and reservoir totals. A resolved atmospheric/surface strip exchanges mass and energy with those summaries. Differentiation, mantle flow, plate-boundary driving and weathering connect interior and surface processes. Climate couples insolation, radiative transfer, clouds and reservoirs. A logged offline stellar-yield calculation supplies the disk composition under proposed decision D6. Later additions require producer attribution on fluxes and finer near-surface tracers. Refinement and demotion reconcile budgets, while long temporal jumps end with orbital and ledger consistency checks.

Prerequisites and interfaces

S1-S5 and S9 address rotation, rheology, nucleation, gravity/versioning and the expansion clock; S12 supplies photochemistry. S19-S20 are additional dependencies for later industrial attribution and local exposure. The planet::env interface exposes thermal, chemical, radiation and nutrient conditions, while biology may return uptake and release events through the common reservoir ledger.

Proposed validation

Source targets include roughly ten thousand resolved atmospheric elements, an accretion segment and a four-billion-year climate experiment; these are unmeasured ambitions. Proposed tests cover differentiation, orbital patterns, convection thresholds, water/carbon budgets, radiation shielding and climate-response controls. Detailed-versus-summary overlap windows, promotion/demotion storms and long-jump re-anchoring should quantify drift and closure error. The seeded composition must retain its generating run's provenance. Climate states and ice-age timings should be reported as conditional outcomes rather than tuned to match a predetermined habitable history.

Research limits

Coarse spatial resolution and extreme time separation may erase feedbacks that control climate or tectonics. An implausible trajectory can expose a model error, but an undesirable outcome is not automatically a bug. Neither stable habitability nor abiogenesis is guaranteed by environmental gradients. A seeded prior stellar yield also does not demonstrate uninterrupted end-to-end chemical history.

Sources: PHASE7.md: Scale contract (what "planetary" means here, numerically); PHASE7.md: Architecture; PHASE7.md: Work packages W1-W7; PHASE7.md: Interface exposed to Phase 8 (life); PHASE7.md: Verification strategy; spec/seams.md: S1-S9, S12, S19, S20. [S1, S3]

Phase 8 Life and ecosystems

Status Detailed proposal. No biological layer is implemented; the current engine demonstrates neither abiogenesis, heredity, ecological evolution nor organisms.

Phase 8 asks whether the proposed chemistry and environments can support persistent, heritable, competing systems. It extends from catalytic networks and compartments to ecosystems and physical bodies, while treating cognition and culture as later questions rather than requiring a prescribed sequence of named organisms.

Proposed mechanisms

Templated reactions would support copying and variation; an observational closure detector identifies catalytic networks without creating them. Amphiphile assembly supplies compartments, and resource uptake, growth, work, heat and waste share an energy ledger. Population summaries retain genotype frequencies and host-parasite correlations; detailed individuals are sampled through deterministic stream addresses. Chemical exchanges and mechanical contacts define food-web edges, decomposition returns material to planetary reservoirs, and regulatory chemistry provides candidate routes to multicellularity. Sensory transduction and motion consume existing physical interfaces. Stable intervals may be compressed; ecological or environmental shocks require a validated detailed treatment.

Prerequisites and interfaces

S10-S12 must add templating, vesicle assembly and photochemistry upstream. The phase reads planetary conditions, radiation/ionization information and material/mechanical state. Its intended writes remain flux and physical interaction events on the shared global clock, clarified by S14. S24 later requires certified population jump intervals rather than allowing the scheduler to invent biological evolution rules.

Proposed validation

Proposed controls compare catalytic and non-catalytic networks, enclosed and free chemistry, supplied and starved resources, and biology-enabled and abiotic reservoirs. Heredity, variation, competition and persistence need separate observables. Population samples should preserve relevant joint statistics and their response to perturbation, not only marginal means. The source also proposes host-parasite adaptation, selective inhibition, trophic patterns and a compressed 500-million-year run. These are hypotheses and scale targets, not measured achievements; complete energy/material accounting and independent chemical benchmarks remain prerequisites.

Research limits

Catalytic closure alone is insufficient to establish life or its origin. A seeded-replicator fallback would change the causal claim and must be disclosed. The specified UV-B-like harvesting band needs physical justification rather than adoption as a photosynthetic benchmark. Rare variation, ecological correlations and failed emergence must survive both model selection and compression, without forcing the desired evolutionary story.

Sources: PHASE8.md: Scale contract (the biological LOD ladder); PHASE8.md: Architecture; PHASE8.md: Work packages W1-W7; PHASE8.md: Interface exposed to Phase 9 (intelligence); PHASE8.md: Verification strategy; PHASE8.md: Risks; spec/seams.md: S10-S14, S24. [S1, S3]

Phase 9 Agency and early civilization

Status Detailed research proposal; agent bodies, learning and civilization are not implemented in the inspected engine.

Extend living lineages into agents that learn, manipulate materials, transmit skills and form persistent settlements. The phase ends around an agricultural and occupational-specialization regime; industry is assigned to later phases. Historical labels identify measured patterns rather than new physical substances or implemented capabilities.

Proposed mechanisms

An agent combines an articulated body, a learned policy and skill memory. Drives derive from resource and energy deficits; actions enter the world as momentum, contact or flux events. Learning estimates budget consequences, while planning extends only as far as predictions remain calibrated. Skills travel through acoustic, visual or material channels. The proposed culture ledger records variant frequency, transmission fidelity and lineage relationships. Cooking is a temperature-time chemical transformation, tool quality follows fracture geometry and wear, and domestication is a response to recorded harvesting or tolerance choices. Outside detailed regions, skill and role distributions replace individual trajectories while retaining lineage-culture correlations.

Prerequisites and interfaces

The plan requires Phase 8 bodies, resource budgets, sensory transduction and lineage records; Phase 5 material properties; Phase 6 contact and fracture behavior; and suitable Phase 7 geography. S13 must supply articulated mechanics. The S14 wording correction retains one global clock. Proposed decision D9 defines when cultural and demographic compression is valid.

Proposed validation

Proposed tests compare viable and depleted resource settings, unfamiliar barriers, communication fidelity above and below a cumulative-learning threshold, and worked versus unworked tool geometry. Cooking-related energy returns, domestication traits and occupational distributions would be measured from the ledgers. Ancestry must remain traceable to preceding lineages. Compressed epochs would be compared with detailed village windows, particularly during shocks. The plan targets at most about 1,000 instantiated agents and roughly 10,000 simulated years; these are unmeasured design ambitions.

Research limits

Learning rewards and specially prepared geography can encode the desired outcome indirectly. Culture and physical tool discovery need controls beyond the absence of named action buttons. Historical transitions must remain contingent, and missing discoveries should be distinguished from software defects. Compressed correlations may lose rare innovations or migration effects.

Sources: PHASE9.md: Scale contract; PHASE9.md: Architecture; PHASE9.md: W1-W7; PHASE9.md: Verification strategy; spec/seams.md: S13-S14. [S1, S3]

Phase 10 Technology and cumulative knowledge

Status Detailed research proposal; technological artifacts, claims ledgers and nested worlds remain unbuilt.

Study agents experimenting with the physical world, building useful artifacts and preserving reproducible knowledge. Technology is proposed as a classification of material structures and their behavior, while science becomes a record of interventions, claims and replications. The current engine demonstrates none of these technological transitions.

Proposed mechanisms

The proposed research loop reuses agent learning with an explicit exploration cost. Instruments map field-dependent outputs into observations and carry calibration errors. A claims ledger stores intervention descriptions, initial-state references and evidence pointers; independent experiments can change a claim's status. Machines combine material records, articulated structures and field interactions, with fuel, useful work and losses recorded together. Physical communication links support packet routing and cultural diffusion. Computing is intended to arise from switching artifacts. A nested simulator would derive child seeds from parent events, include child outcomes in parent provenance, obey a recursion guard and consume an in-world hardware budget.

Prerequisites and interfaces

Phase 9 learning and cultural transmission, Phase 5 materials, Phase 6 mechanics and Phase 7 environments must first be validated. S15 extends electromagnetic modeling beyond magnetostatics to circuit and propagation regimes. S16 and proposed decision D10 define nested replay. Ownership-transfer records required by later economic phases are also owed upstream through S18.

Proposed validation

The plan proposes heat-engine energy and entropy checks, grid responses to changing loads, signal propagation, instrument recovery of known world parameters, and replay of stored experiments. Biological interventions would be tested through dose-response and reservoir changes. Nested runs must reproduce their own results and reject prohibited recursion. Communication effects would be compared with links enabled and disabled. Device and adoption classifiers need perturbation tests. The proposed 100-year industrial scenario and nested-cost budget are targets without integrated measurements.

Research limits

Material trend models may not support useful semiconductor behavior, and rare invention chains may be computationally inaccessible. Directly invoking a child engine does not establish equivalence to computation performed by simulated hardware. Required demonstration milestones also conflict with language allowing technologies to be skipped; these conditions need reconciliation before preregistration.

Sources: PHASE10.md: Scale contract; PHASE10.md: Architecture; PHASE10.md: W1-W7; PHASE10.md: Verification strategy; spec/seams.md: S15-S16 and S18. [S1, S3]

Phase 11 People and society

Status Detailed research proposal; social institutions and populations are not implemented in the current baseline.

Represent persistent collective patterns through interactions among agents, institutions, communication and resource use. The proposed social layer adds interpretable records over earlier mechanisms. It should support controlled comparisons of social outcomes without treating model categories as validated descriptions of real populations or clinical conditions.

Proposed mechanisms

Sanction graphs record who enforces which expectations and at what cost. Agents encounter those consequences through existing perception and learning channels; the social layer must not directly rewrite policy weights. Training changes skills through time and resource expenditure, while occupational distributions respond to exchange opportunities. Language-like variation is represented through transmission fidelity and mutual intelligibility across geography. Coalitions, membership events and shared claim complexes support persistent group records. Health-related observables combine bodily budgets, exposure and policy-calibration measures. Conflict consumes logistical resources and depends on available artifacts. Stable regions retain compressed graphs and distributions, while institutional shocks require more detailed evolution.

Prerequisites and interfaces

The phase requires Phase 8 physiology and exposure mechanisms, Phase 9 agents and cultural records, and Phase 10 artifacts, claims and communication networks. S17 fixes the interface wording so norms act through sensed environments rather than privileged policy edits. Proposed decision D11 extends the validity conditions for demographic compression to institutions and social graphs.

Proposed validation

Proposed experiments include shared-resource communities, language transects with barriers, matched educational cohorts and exposure interventions with delayed recovery. Enforcement expenditure, resource persistence, learning curves and communication fidelity provide measurable outputs. Group records must remain consistent through membership turnover. Artifact-limited conflict and detailed-versus-compressed city comparisons would test additional interfaces. The proposed scale is about 10,000 compressed agents with at most 1,000 instantiated; runtime and model-accuracy measurements are still absent.

Research limits

The same aggregate pattern can arise through many incompatible mechanisms. A pair of communities producing different outcomes is not by itself evidence of realistic governance, and requiring predetermined contrasting outcomes can introduce selection bias. Policy-calibration distributions cannot be equated with human mental health without independent evidence. Social compression must preserve the correlations that determine responses to shocks.

Sources: PHASE11.md: Scale contract; PHASE11.md: Architecture and Data rules; PHASE11.md: W1-W7; PHASE11.md: Verification strategy; spec/seams.md: S17. [S1, S3]

Phase 12 Money power and political organization

Status Detailed research proposal; ownership, finance and governance systems are not implemented in the inspected engine.

Connect ownership, transferable claims, credit and enforcement to the world's existing physical flows. Monetary and political categories are proposed as interpretations of event networks and record structures. Their accounting must remain explicit, while behavioral and institutional outcomes remain hypotheses to be evaluated under specified conditions.

Proposed mechanisms

Ownership becomes an engine relation expressed through transfers and holdings that survive changes in resolution. Obligation records specify counterparties, amounts and terms; agents estimate repayment prospects, and circulation can produce a commonly used denomination. Banking coalitions hold assets and issue claims with differing maturities. Extraction and succession mechanisms operate through enforcement, communication and transfer events. Verification of claim chains consumes simulated computing resources, with disputes creating competing rule histories. Propaganda and correction are modeled through claim transmission, replication status and audit capacity. Prices and regime labels are intended as derived measurements rather than primitive state variables or equilibrium targets.

Prerequisites and interfaces

Phase 11 provides enforcement and social graphs, while Phase 10 provides records, artifacts, networks and reproducibility mechanisms. S18 requires ownership-transfer events and compressed holdings upstream. S16/D10 supplies the proposed verification and nested-computation contract. Proposed decision D12 specifies the validity domain for compressing record graphs and prediction distributions.

Proposed validation

The plan calls for exact holdings preservation, reconstructible exchange statistics, and explicit separation of nominal records from physical resources. Bank-run tests compare the timing of expectation changes with asset depletion. Additional experiments examine supply shocks, convertibility changes, verification forks, institutional costs, mobility and information correction. Both physical and accounting budgets must close. Compressed states require checks around tipping windows, where averaging may reverse causal ordering. The planned population of roughly 100,000 claim records per region has no measured performance result.

Research limits

Record conservation cannot establish that monetary changes leave production unchanged. Prescribed regime advantages, inequality tails or misinformation rankings are conditional hypotheses rather than universal economic laws. Excluding price variables or named regimes does not remove behavioral assumptions. Independent scenarios and negative results are needed to avoid fitting the desired political narrative.

Sources: PHASE12.md: Scale contract; PHASE12.md: Architecture and Data rules; PHASE12.md: W1-W7; PHASE12.md: Verification strategy; spec/seams.md: S18. [S1, S3]

Phase 13 Infrastructure industry and externalities

Status Detailed research proposal; infrastructure, industrial transport and liability mechanisms have not been built.

Link built infrastructure and industrial production to resource depletion, maintenance, exposure and responsibility. The phase proposes a traceable feedback loop from physical production through transport and consequences to collective action, while retaining a strict distinction between measured exposure and an established effect.

Proposed mechanisms

Structures are owned artifacts with accumulated load, damage and maintenance histories. Terrain influences construction costs and feasible transport networks. Agent routing and learned coordination produce traffic conventions; freight is constrained by capacity, temperature and spoilage. Regional accounts integrate food, water, energy, waste and recycling. Sparse producer tags accompany industrial fluxes, while local tracer transport resolves concentrations near settlements. Exposure integrates location, duration and concentration; harm records require an applicable response model. Liability records connect source signatures, chemical form, transport and consequences. Network failures redistribute loads, and orbital debris uses aggregate shells with detailed conjunction windows.

Prerequisites and interfaces

Validated materials, continuum mechanics, planetary transport, biological response and ownership records are prerequisites. S19 adds producer attribution to upstream flux events; S20 adds a finer near-surface advection, diffusion and deposition channel. S2 supplies slow deformation, S13 moving articulated machinery and S7 resolution discipline. Proposed decision D13 governs compression of attribution, exposure and liability records.

Proposed validation

Proposed tests examine terrain-dependent routes, maintenance-related failure ordering, convention-boundary collisions, water budgets, thermal limits on food transport and recycling energy comparisons. The principal integrated test follows a liability citation across composition, transport and response records. Emission interventions must preserve physical recovery delays. Matched storms on differently maintained networks test vulnerability separately from hazard. Unmodeled exposure effects must remain explicitly unknown. Targets include about 10,000 structures and a bubble-local tracer grid; all associated cost figures remain unmeasured.

Research limits

Producer tagging can lose attribution under mixing or compression, and a coarse transport field can misrepresent local exposure. Citation completeness does not validate a causal response model or determine legal responsibility. Passive tracers omit reactive transport unless coupling is supplied elsewhere. Hazard and cascade outcomes require ensemble and resolution studies rather than a single selected event.

Sources: PHASE13.md: Scale contract; PHASE13.md: Architecture and Data rules; PHASE13.md: W1-W7; PHASE13.md: Verification strategy; spec/seams.md: S19-S20. [S1, S3]

Phase 14 Perception memory and level of detail

Status Detailed research proposal; only the simpler probe and restricted quiet-cell precursors exist in the baseline.

Replace the external observer fixture with observation requests generated by simulated agents and records, while defining what numerical detail can be reduced. The plan separates physical state transitions from knowledge about them. Perception-dependent computation must not be mistaken for a demonstrated theory that observation creates physical existence.

Proposed mechanisms

Gaze combines position, orientation, light and occlusion, with instruments and archive consultations extending observation. Frame allocation derives from simulated scene complexity and agent budgets, never host timing. Knowledge records carry values, resolution, confidence and provenance; remembered or inferred values must remain distinguishable from observations. Testimony propagates through communication and intelligibility constraints, and reconciliation retains source histories instead of simply averaging beliefs. A governor chooses detail using observer overlap, causal proximity and persistent records. Hysteresis discourages oscillation, while certified summaries govern unobserved intervals. Physical and epistemic records are compared through explicit consistency rules.

Prerequisites and interfaces

The plan consumes Phase 13 infrastructure and attributed ledgers, Phase 9 cognitive budgets, Phase 10 instruments and networks, and Phase 11 communication models. S21 records a corrected observation-source contract, but its mechanics remain future work. S22 requires deterministic allocation traces in replay. Proposed decisions D14 and D15 formalize two-level observation and frame-budget determinism.

Proposed validation

A gaze-derived map must match the existing probe fixture where their inputs agree. Proposed tests inject host-clock jitter, unsupported belief detail, conflicting testimony and resolution oscillation. Archived evidence should constrain later reconstructed beliefs, and unobserved events must remain discoverable through preserved causal records. Detailed and reduced runs would be compared on physical and epistemic observables using a metric fixed before optimization. The plan targets at least 40% lower tick cost, including governor overhead; this saving has not been measured.

Research limits

Perceptual relevance may omit unseen causes that later become consequential. Exact restoration applies only to information actually retained; it cannot recover discarded detail without assumptions. The comparison metric must cover rare events and correlations, not only aggregate beliefs. Consistency with the baseline collapse rule does not independently validate that rule's physical interpretation.

Sources: PHASE14.md: Scale contract; PHASE14.md: Architecture and Data rules; PHASE14.md: W1-W7; PHASE14.md: Verification strategy; spec/seams.md: S21-S22. [S1, S3]

Phase 15 Runtime scheduling and scalable execution

Status Detailed research proposal; the unified runtime, GPU execution and whole-world certificates are unimplemented.

Reduce computational cost while preserving the declared semantics and measured accuracy of the model hierarchy. The proposed runtime changes storage, scheduling, batching and device placement; it should not privately invent missing physical or social closure rules. Performance claims would concern complete validated workloads.

Proposed mechanisms

A unified data-oriented store organizes state into archetype segments with index-based references and explicit snapshots. Tiered scheduling combines region resolution with each layer's permitted update interval. Every skipped interval requires a layer-owned certificate and a defined response when its conditions fail. The proposed device split places cell-based physical calculations on GPUs and cognition and record processing on CPUs. Ordered causal channels preserve events, while selected aggregate or perception channels may incur declared delay. Fixed-point or interval representations expose quantization choices. Run records would include build identities, schedule information, throughput, memory traffic, duty cycles and error-budget expenditure.

Prerequisites and interfaces

The full Phase 14 perception and reduction contract must first be implemented and tested. S23 adds a runtime-build fingerprint and distinguishes replay within one build from comparison across builds. S24 requires interval-certified summaries from materials, biology, technology, society and monetary records. S16 covers child runs; proposed decision D16 specifies runtime equivalence rather than silently expanding existing reproducibility claims.

Proposed validation

The plan retains earlier acceptance suites and compares transformed state against reference calculations. Proposed stress tests perturb queue order, transfers, partitioning and host scheduling. Snapshot round trips, certificate violations, event accounting and both device channels receive separate checks. A hardware-named performance record would measure complete runs and regression variability. Target throughput multipliers and an interactive world with roughly 10,000 agents on one reference card are ambitions, not demonstrated capacities.

Research limits

A fixed reduction order does not alone guarantee identity across hardware, compilers and numerical libraries. Quantization bit width bounds representation error only after units, ranges and accumulated propagation are established. Passing a finite corpus does not prove universal equivalence. Coarse biological or institutional dynamics remain scientific modeling work that a scheduler cannot supply.

Sources: PHASE15.md: Scale contract; PHASE15.md: Architecture and Data rules; PHASE15.md: W1-W7; PHASE15.md: Verification strategy; spec/seams.md: S23-S24. [S1, S3]

Phase 16 Integrated emergence and the full experimental loop

Status Detailed terminal-phase proposal; the integrated harness and complete simulated histories are not implemented.

Evaluate a complete, persistent history across the preceding mechanisms using an integration harness rather than another physical layer. The proposed outcome includes endogenous innovation, in-world computation and playable historical states. It remains a program-level ambition; no complete quantum-to-civilization run currently exists.

Proposed mechanisms

A committed initialization, pinned model and recorded events would define each run. Read-only milestone detectors are specified before execution and query a shared event ontology with causal and ordering predicates. Reports would combine milestone timing, structural differences among histories, resource costs and sustained novelty. The proposed ladder includes a communication mesh expanding beyond political boundaries and a depth-one nested world commissioned by agents. Child results must enter the parent claims record and influence subsequent choices. Historical access would reconstruct a selected prefix through certified summaries, then continue detailed evolution where observation requires it.

Prerequisites and interfaces

Every preceding phase must have implemented mechanisms and validated interfaces. S25 requires a machine-readable event ontology derived from interface contracts and event-kind registries. S23 and S24 supply build identity and certified coarse intervals. D6 initial-condition choices and D10 nested provenance must be disclosed; proposed decision D17 addresses how the project operationalizes unscripted emergence.

Proposed validation

The plan proposes at least three gating seeds, detector negative controls, causal-ancestry diversity measurements, replay checks and an adversarial recursion test. In-world replication should expose an error in a nested model. Additional targets include opening a historical state within 60 seconds, continued invention over another simulated century, and per-era time and storage accounting. These are future acceptance requirements with blank measurement fields, not completed demonstrations or established computational budgets.

Research limits

Frozen detectors, noun filters and a few diverse seeds cannot prove the absence of scripting or general emergence. Engine rules, reward functions and initialized geography can encode outcomes indirectly. Requiring every seed to realize a fixed historical ladder conflicts with genuine contingency. Failed emergence may reveal model boundaries or unsuitable initial conditions rather than a programming bug. Cross-machine identity remains conditional on a demonstrated numerical contract.

Sources: PHASE16.md: Scale contract; PHASE16.md: Architecture and Data rules; PHASE16.md: W1-W6; PHASE16.md: Verification strategy; spec/seams.md: S25. [S1, S3]

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Project sources and evidence index

[S1] SimWorld roadmap and phase specifications. ROADMAP.md and PHASE0.md through PHASE16.md at commit 4134abd, 29 October 2026. All seventeen detailed plans are available. Sections 9-11 summarize the program; Appendix B gives each phase's mechanisms, prerequisites, proposed validation and research limits.

[S2] SimWorld first build contract. BUILD0.md. Relevant sections: Environment; What ships; Decisions; Physics build 0.1; Engine architecture; Physics flow per tick; verification and milestone acceptance criteria.

[S3] SimWorld specification records. spec/constants.md; spec/decisions.md; spec/interfaces.md; spec/surprises.md; spec/seams.md. Decisions and interfaces carry the physics_build 0.1 freeze date of 28 October 2026. The surprises and seams ledgers are records of known limitations and planned changes.

[S4] SimWorld implementation at commit 4134abd. The sim package contains the reduced model described in Sections 3-4. Compared with the original audit snapshot, atoms.py, bind.py, log.py, network.py, quanta.py and state.py have comment/docstring corrections; all 19 module executable ASTs match after docstrings are excluded. The archive preserves both source versions. [S8]

[S5] SimWorld verification suite. tests/test_rng.py; test_log.py; test_replay.py; test_m2_physics.py; test_m2_jump.py; test_m3_nuclei.py; test_m4_atoms.py; test_physics_bands.py; physics_bands.toml; tests/oracle/. The frozen anchors and acceptance criteria are part of the model-development evidence, not wholly independent physical validation.

[S6] Archived runs and prototype reports. Historical M3/M4 summary, compressed log and state-hash records are retained under baseline_source/_m3run/ and baseline_source/_m4run/. Prototype code and reports are in spikes/a_quant_cost/ and spikes/b_shells/ in both source snapshots. These measurements describe their individual workloads and retain their original provenance.

[S7] Original local audit evidence generated 29 October 2026. research/audit_tests.txt and audit_tests.xml; audit_diagnostics.py and audit_diagnostics.json; fresh_m4/; implementation_audit.txt; literature.json; and the preserved baseline source manifest. These records retain the original measurements. They are distinct from the later CI verification and v1.1 phase profiles.

[S8] Publication status audit, 29 October 2026. research/publication_status.json identifies source commit 4134abd, CI run 40857373977, successful Windows and Ubuntu jobs, Pages run 40857374197, 17 plan fingerprints, the 25-entry seam ledger, and the executable-AST comparison. These public records support the Current project status page; they do not replace the original physical diagnostics.

Evidence locations for the principal findings

Finding	Source location
Dictionary/list quant storage	sim/quanta.py, QuantStore.__init__ and add_superposed
Reverse nuclear channel omissions and formula	sim/network.py, _rates; sim/bind.py, channel
Fully neutral absorbing state	sim/atoms.py, step; audit_diagnostics.json, atomic_absorbing_diagnostic
Random-prefix reuse	sim/log.py, EventContext.draws; audit_diagnostics.json, rng_reuse_diagnostic
Fresh equality and replay	audit_diagnostics.json, snapshot_comparison and runs; fresh_m4/hashes.json

Table 9. Function-level locators remain usable in the archived source even when line numbers change in later development.